

Computer Networking

Master Exam Study Notes

Combined Modules – high-yield notes, comparisons, diagrams, formulas, and question bank

Exam Focus

Exam strategy. These notes combine all three Networking modules into one study bank because the examination draws from the complete syllabus. The supplied lecture PDFs are the primary technical source; the manually prepared topic checklist is used as a coverage checklist. Prioritize definitions, layer responsibilities, mechanisms, labelled diagrams, comparisons, formulas, and answer structures that can be expanded into 5–10 marks.

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1. Networking Fundamentals

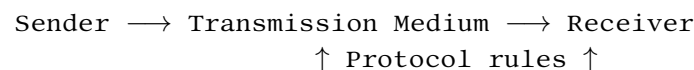
1.1 What is Data Communication?

Data communication is the exchange of data between two devices through a transmission medium. Effective data communication is characterized by **delivery, accuracy, timeliness, and jitter**.

Characteristic	Meaning
Delivery	Data should reach the intended destination.
Accuracy	Data should arrive without unacceptable alteration.
Timeliness	Data should arrive when required; excessive delay can make data useless.
Jitter	Variation in packet arrival time, especially important for real-time traffic.

1.2 Five Components of Data Communication

1. **Message** – the information being communicated.
2. **Sender** – the device that sends the message.
3. **Receiver** – the device that receives the message.
4. **Transmission medium** – the physical path through which the signal travels.
5. **Protocol** – the rules governing communication.



1.3 Data Flow

Mode	Direction	Example / idea
Simplex	One direction only	One side transmits; the other only receives.
Half-duplex	Both directions, but not simultaneously	Walkie-talkie style communication.
Full-duplex	Both directions simultaneously	Telephone communication.

1.4 What is a Network?

A computer network is an interconnection of devices capable of communicating with one another. Typical devices include hosts, switches, routers, and modems.

1.5 Network Criteria

- **Performance** – commonly discussed using throughput and delay.
- **Reliability** – how often failures occur and how quickly the system recovers.
- **Security** – protection against unauthorized access, damage, and data loss.

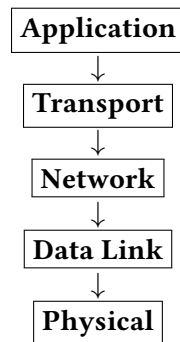
Remember

Throughput and delay are not the same. Throughput is the effective data rate; delay is the time involved in moving data. Congestion can increase delay even when the network has high nominal capacity.

2. Protocol Layering, OSI and TCP/IP

2.1 Why Do We Use Layers?

Networking is complex because communication involves applications, end-to-end transport, logical addressing, routing, link-level delivery, and physical transmission. Layering divides this complexity into manageable responsibilities.

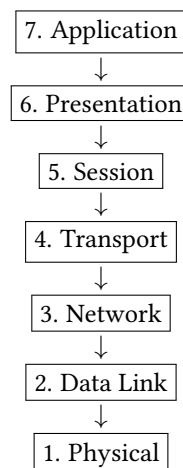


2.2 TCP/IP Protocol Suite

Layer	PDU / logical unit	Important examples / role
Application	Message / data	HTTP, SMTP, FTP, TELNET, SSH, SNMP, DNS, IGMP. Application-level communication.
Transport	Segment / user datagram	TCP and UDP; process-to-process/end-to-end transport.
Network	Datagram / packet	IP; host-to-host delivery and forwarding/routing.
Data Link	Frame	Node-to-node delivery across one link; framing and MAC access.
Physical	Bits	Carries bits as electrical, optical, or radio signals.

2.3 OSI Reference Model

The OSI reference model contains seven layers.



Layer	Exam memory point
Application	Network services visible to applications.
Presentation	Data representation, translation, encryption/compression concepts.
Session	Establishes, manages and terminates logical sessions.
Transport	End-to-end transport between processes.
Network	Logical addressing and routing between networks.
Data Link	Framing and node-to-node delivery over one link.
Physical	Bit and signal transmission.

2.4 PDU Cheat Sheet

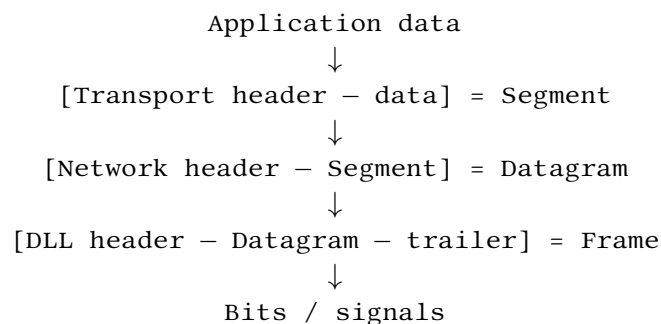
Layer	Common PDU
Application	Message / data
Transport	Segment (TCP) / user datagram (UDP)
Network	Datagram / packet
Data Link	Frame
Physical	Bits

Remember

Memorize the chain: Message → Segment/User Datagram → Datagram/Packet → Frame → Bits.

2.5 Encapsulation and Decapsulation

At the sender, application data moves down the protocol stack. Each layer adds its own control information. At the receiver, the reverse process occurs.



2.6 Protocols Associated with the Network Stack

Protocol	Role
TCP	Connection-oriented transport; flow, error and congestion control.
UDP	Connectionless transport; independent user datagrams without TCP's flow/error/congestion mechanisms.
IP	Network-layer datagram format, logical addressing and forwarding.
ICMP	Reports network-layer delivery/routing problems.
IGMP	Supports multicast/group membership.
DHCP	Helps a host obtain a network-layer address.
ARP	Resolves a known IP address to a local link/MAC address.

2.7 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Setup	Logical connection established first	No prior connection setup
Flow control	Yes	No
Error control	Yes	No
Congestion control	Yes	No
Data model	Byte stream / segments	Independent user datagrams

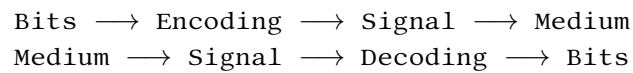
Exam Focus

Likely 5-mark question: Explain the OSI/TCP-IP models and identify the PDU at each layer. A strong answer should include a layer diagram and one or two protocols at important layers.

3. Physical Layer

3.1 Physical Layer Responsibility

The Physical Layer is responsible for carrying individual bits across a link. The medium carries electrical, optical, or radio signals; the physical layer defines how the bit representation is transformed into a transmissible signal.



3.2 Data and Signals

Term	Meaning
Analog data	Continuous information.
Digital data	Information represented using discrete values.
Analog signal	Continuously varying signal.
Digital signal	Signal represented using a limited set of defined levels.

3.3 Sine-Wave Parameters

Parameter	Meaning	Relationship / unit
Amplitude	Peak strength of the signal.	Often volts for electrical signals.
Period T	Time for one complete cycle.	seconds; $T = 1/f$
Frequency f	Cycles per second.	Hz; $f = 1/T$
Phase	Position of waveform relative to time zero.	degrees/radians
Wavelength λ	Distance travelled during one period.	$\lambda = v/f$

3.4 Bandwidth

For a composite signal, bandwidth is the difference between its highest and lowest frequencies.

$$\text{Bandwidth} = f_{\text{high}} - f_{\text{low}}$$

3.5 Bit Rate and Bit Length

- **Bit rate** is the number of bits transmitted per second, measured in bits per second (bps).
- **Bit length** is the physical distance occupied by one bit on the transmission medium.

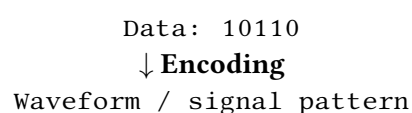
3.6 Guided and Unguided Media

Category	Examples	Core idea
Guided media	Twisted pair, coaxial cable, optical fiber	Signal follows a physical path.
Unguided media	Radio, microwave, infrared	Signal propagates through free space/air.

3.7 Encoding and Decoding – EASY VERSION

Encoding means converting data bits into a signal representation suitable for transmission.

Decoding means interpreting the received signal and recovering the original bit representation.



↓ **Transmission**
 Received waveform / signal
 ↓ **Decoding**
 Data: 10110

Remember

Do not overcomplicate this. The core idea is simply **bits** → **signal** at the sender and **signal** → **bits** at the receiver. The exact waveform depends on the encoding scheme.

4. Data Link Layer – Deep Dive

4.1 Node-to-Node Communication

The Data Link Layer handles communication between adjacent nodes over a single link. Hosts and routers are nodes, while the networks/links between them form the path across which data travels.

Host A ↔ Router 1 ↔ Router 2 ↔ Host B
 Link 1 Link 2 Link 3

The Network Layer selects the next hop; the Data Link Layer moves the Network Layer datagram across the selected link.

4.2 Data Link Layer PDU

The Network Layer gives the Data Link Layer a packet/datagram as data. The Data Link Layer encapsulates it into a **frame**.

[Frame Header – Network-layer packet/datagram – Frame Trailer]

The header carries control and link-address information; the trailer commonly carries an error-detection value such as a checksum/CRC.

4.3 Services Offered by the Data Link Layer

Service	Meaning
Unacknowledged connectionless	Independent frames are sent without acknowledgments. Ethernet is given as an example in the supplied material.
Acknowledged connectionless	No logical connection, but each frame is individually acknowledged.
Acknowledged connection-oriented	A logical connection is established; frames are numbered and acknowledged, with ordered delivery according to the lecture description.
Framing	Divides the bit stream into identifiable frames and enables frame-level checking.
Flow control	Prevents a fast sender from overwhelming a slow receiver.
Error control	Uses feedback, acknowledgments and timers/retransmission concepts to deal with corrupted or lost frames.
Congestion control	Addresses overload on a link/network where excessive traffic can cause loss and delay.

4.4 Point-to-Point and Broadcast Links

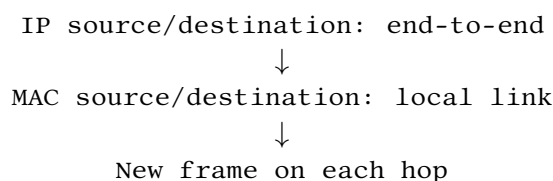
Point-to-point	Dedicated link between two devices.
Broadcast	Shared medium used by multiple devices; access control becomes important.

4.5 DLC and MAC Sublayers

DLC (Data Link Control)	Handles link-control functions such as framing, flow control and error control.
MAC (Media Access Control)	Handles access to a shared/broadcast medium.

4.6 Link-Layer Addressing

An IP datagram carries logical source and destination addresses. The frame used on each individual link carries link-layer/MAC addresses.



Do Not Confuse

Do not confuse them. The logical IP destination remains the destination of the datagram. The MAC/link addresses are concerned with the current local hop and can change at every router-to-router or host-to-router link.

4.7 Unicast, Multicast and Broadcast

Type	Pattern	Link scope
Unicast	One-to-one	One destination entity.
Multicast	One-to-many	A selected group of entities on the link.
Broadcast	One-to-all	All relevant entities on the local link.

5. Framing, Flow Control and Error Control

5.1 Why Framing is Needed

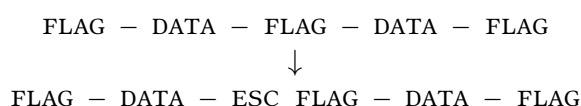
The Physical Layer provides a stream of bits. The Data Link Layer needs to determine where one unit ends and the next begins. **Framing** divides the bit stream into manageable frames.

5.2 Framing Methods

Method	Working
Byte count	A field specifies the number of bytes in the frame. The receiver uses this count to locate the boundary.
Flag bytes + byte stuffing	Special flag bytes delimit the frame. If a flag-like value appears in data, an escape byte is inserted; the receiver removes it.
Flag bits + bit stuffing	After five consecutive 1s in the data, the sender inserts a 0. The receiver removes the stuffed 0.

5.3 Byte Stuffing – Simple Example

Suppose FLAG marks the start/end of a frame and the data itself contains FLAG. The sender inserts an escape before the data occurrence.



The receiver recognizes that ESC FLAG is data, not the end of the frame, and removes the escape.

5.4 Bit Stuffing

If a flag contains a run of 1s, the sender prevents accidental flag detection by inserting a 0 after five consecutive 1s in the data.

Data: 1111101
Stuffed: 11111001

5.5 Flow Control

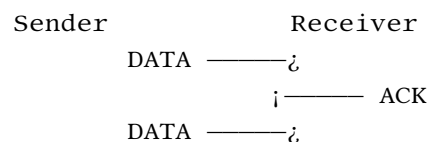
Flow control regulates transmission so a sender does not overwhelm a receiver.

Feedback-based	Receiver feedback influences how much/how fast the sender transmits.
Rate-based	The protocol directly controls the sender's transmission rate.

5.6 Stop-and-Wait

Stop-and-wait uses one outstanding frame at a time.

1. Sender receives a packet from the Network Layer.
2. Sender creates a frame, saves a copy, starts a timer, and sends it.
3. Sender waits for an acknowledgment.
4. Correct ACK → stop timer, discard saved copy, send next frame.
5. Timeout → retransmit the saved frame.
6. Corrupted/invalid ACK → discard it and continue waiting.
7. Receiver checks the frame; if correct, it delivers the data upward and sends an ACK.



5.7 Sequence Numbers

Sequence numbers distinguish a new frame from a retransmission of an old frame. A simple alternating sequence uses 0, 1, 0, 1, ...

5.8 Piggybacking

Piggybacking attaches an acknowledgment to an outgoing data frame instead of sending a separate ACK frame immediately. This improves channel utilization because the ACK travels with useful data.

5.9 Sliding Window

Sliding-window protocols allow multiple frames to be outstanding before acknowledgments arrive. This improves link utilization on links with significant propagation delay.

[ACKed][ACKed][SENT][SENT][SENT][AVAILABLE][AVAILABLE]
 ↑ sending window moves →

5.10 Go-Back-N vs Selective Repeat

Feature	Go-Back-N	Selective Repeat
Receiver after error	Discards/does not accept later out-of-order frames in the supplied description.	Can accept and buffer later correct frames.
Retransmission	Retransmits the missing/damaged frame and following unacknowledged frames.	Retransmits only the required missing/damaged frame(s).
Efficiency	Lower when errors occur.	Higher when errors are isolated.
Complexity	Simpler.	More complex due to buffering/state.

5.11 Errors

Single-bit error	One bit changes from 0 to 1 or 1 to 0.
Burst error	Two or more bits in the data unit are altered.

5.12 Redundancy and Coding

The sender adds redundant bits so the receiver can detect or sometimes correct errors.

For block coding:

$$n = k + r$$

where k is the number of data bits, r is the number of redundant bits, and n is the codeword length.

5.13 Hamming Distance

The Hamming distance between two equal-length words is the number of bit positions in which they differ.

$$d_{\min} = s + 1$$

A code with minimum Hamming distance $d_{\min}$ can detect up to $s = d_{\min} - 1$ errors under the relationship given in the supplied material.

5.14 Parity Check

A parity bit is added to the data. In an even-parity scheme, the parity bit is chosen so that the total number of 1s becomes even.

5.15 Cyclic Redundancy Check (CRC)

1. Sender appends $n - k$ zeros to the dataword.
2. The resulting pattern is divided by a predefined generator using modulo-2 division.
3. The remainder is appended to form the transmitted codeword.
4. Receiver divides the received codeword by the same generator.
5. A zero remainder indicates no error detected by this CRC check; a nonzero remainder indicates an error.

Exam Focus

Likely 5–10 mark area: Explain framing, flow control and error control. For a longer answer, include stop-and-wait, sliding window, Go-Back-N/Selective Repeat, and CRC.

6. Media Access Control and Random Access

6.1 Why MAC is Needed

A broadcast link is shared by several devices. If two devices transmit at the same time, their signals can interfere. The MAC sublayer therefore controls access to the shared channel.

6.2 Channel Allocation Questions

- Is time continuous or divided into slots?
- Does a station sense the carrier before transmitting?
- What happens if a collision occurs?
- How does a station know whether transmission succeeded?

6.3 Random Access Protocol

In random access, stations compete for the channel rather than receiving permanently assigned transmission opportunities. Collisions are possible and must be handled.

6.4 Pure ALOHA

Pure ALOHA is a simple random-access protocol.

1. A station transmits whenever it has a frame.
2. If another station transmits during the vulnerable interval, a collision may occur.
3. The sender waits for an acknowledgment.
4. If the ACK does not arrive, the sender waits a random backoff time and retransmits.
5. After too many unsuccessful attempts, the frame may be discarded/handled later according to the protocol.

6.5 Vulnerable Time

If a frame requires T_{fr} seconds to transmit, Pure ALOHA has a vulnerable period of:

$$2T_{fr}$$

6.6 Pure ALOHA Throughput

For offered load G , the throughput relation is:

$$S = Ge^{-2G}$$

Maximum throughput is approximately:

$$S_{\max} \approx 0.184$$

at $G = 1/2$.

6.7 Slotted ALOHA

Time is divided into synchronized slots of one frame-transmission time. A station can begin transmitting only at the beginning of a slot.

$$\text{Vulnerable time} = T_{fr}$$

$$S = Ge^{-G}$$

$$S_{\max} \approx 0.368$$

at $G = 1$.

Feature	Pure ALOHA	Slotted ALOHA
Start time	Any time	Beginning of a synchronized slot
Vulnerable time	$2T_{fr}$	T_{fr}
Maximum throughput	≈ 0.184	≈ 0.368
Main improvement	None	Synchronization reduces the collision window.

6.8 CSMA

CSMA stands for **Carrier Sense Multiple Access**. The basic principle is:

Sense before transmit / Listen before talk

A station checks whether the medium is busy before transmitting. CSMA reduces collisions, but it cannot eliminate them because a signal takes time to propagate.

6.9 Persistence Methods

Method	Channel idle	Channel busy
1-persistent	Transmit immediately.	Keep sensing until idle, then transmit.
Non-persistent	Transmit immediately.	Wait a random time, then sense again.
p-persistent	With probability p , transmit; with probability $q = 1 - p$, defer to the next slot.	Follows the slotted channel procedure.

6.10 CSMA/CD

CSMA/CD adds **collision detection**.

1. Station senses the medium.
2. If the medium is idle, it starts transmitting.
3. While transmitting, it monitors the medium for a collision.
4. If a collision is detected, transmission is aborted.
5. The station follows the retransmission/backoff procedure.
6. It later attempts transmission again.

```
Sense → Transmit → Detect collision?
                        ↓ yes
                Abort → Backoff → Retry
```

6.11 Why Can Collisions Still Happen?

Suppose A begins transmitting. Because of propagation delay, another station C may still sense the medium as idle before A's signal reaches C. C then transmits and the signals collide.

Remember

CSMA reduces collisions; it does not magically eliminate propagation delay.

6.12 CSMA/CA

CSMA/CA is associated with wireless networks. The supplied material lists collision-avoidance mechanisms including inter-frame space, contention window, and acknowledgments.

Exam Focus

High-yield sequence: Pure ALOHA → vulnerable time → throughput → Slotted ALOHA → CSMA → persistence methods → CSMA/CD → CSMA/CA.

7. ARP – Address Resolution Protocol

7.1 Purpose of ARP

ARP resolves a known **IP address** into the corresponding **local link/MAC address** needed for frame delivery.

```

Known: Destination IP
      ↓ ARP
Needed: Destination MAC / link address
      ↓
Build local Ethernet/link-layer frame
  
```

Do Not Confuse

Direction to remember: ARP primarily performs **IP address** → **MAC/link address** resolution. Do not write MAC → IP as the primary operation.

7.2 ARP Operation

1. Host A wants to send an IP datagram to an IP destination on its local link.
2. Host A checks its ARP cache for the required IP-to-MAC mapping.
3. If the mapping is absent, Host A broadcasts an ARP request.
4. The device owning the target IP recognizes the request.
5. It sends an ARP reply containing its MAC/link address.
6. Host A stores the mapping in its ARP cache.
7. Host A can now construct the link-layer frame using the resolved MAC address.

```

Host A                Local Link                Host B
IP: 192.168.1.10      IP: 192.168.1.20
A – ARP Request: Who has 192.168.1.20? → B
A ← ARP Reply: 192.168.1.20 is at MAC-B ← B
A ===== Data frame addressed to MAC-B =====> B
  
```

7.3 ARP Request vs Reply

ARP Request	ARP Reply
Broadcast on the local link.	Response identifies the target's MAC address.
Contains sender hardware/IP information and target IP.	Returns the target hardware/link address.
The supplied material uses FF:FF:FF:FF:FF:FF as the broadcast target hardware address.	Usually sent back to the requesting host.

7.4 ARP Cache

An ARP cache stores resolved IP-to-MAC mappings so the host does not need to broadcast an ARP request for every packet. Entries remain for a limited residence time and eventually expire.

7.5 ARP Packet Format

Field	Size	Purpose
HTYPE	16 bits	Hardware/network hardware type.
PTYPE	16 bits	Network-layer protocol type.
HLEN	8 bits	Hardware address length.
PLEN	8 bits	Protocol address length.
OPER	16 bits	Operation: request or reply.
SHA	Variable	Sender hardware/MAC address.
SPA	Variable	Sender protocol/IP address.
THA	Variable	Target hardware/MAC address.
TPA	Variable	Target protocol/IP address.

Exam Focus

Very important: If ARP appears as a 10-mark question, draw the request/reply operation, explain the cache, and reproduce the packet-format fields.

8. Multiplexing – Very Important

8.1 What is Multiplexing?

Multiplexing allows multiple signals to share a single data link. It is useful when the link has more capacity than one source requires.



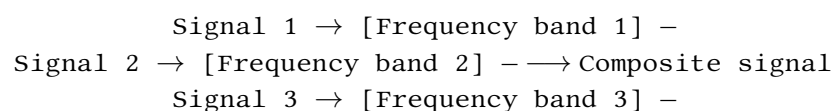
At the receiver, a **DEMUX** separates the combined transmission into the individual output signals.

8.2 Three Basic Techniques

Technique	Resource division	Main idea
FDM	Frequency	Each signal receives a different frequency band.
TDM	Time	Signals use different time slots.
WDM	Optical wavelength	Optical signals use different wavelengths on one fiber.

8.3 FDM – Frequency-Division Multiplexing

FDM is an analog multiplexing technique in which the available bandwidth is divided into frequency channels. Each source modulates a different carrier frequency, and the resulting signals are combined.



Guard Bands

Guard bands are unused frequency gaps placed between adjacent channels to reduce overlap/interference.

8.4 FDM Example

Suppose three voice channels each require 4 kHz and the available link spans 20–32 kHz.

Channel 1	20–24 kHz
Channel 2	24–28 kHz
Channel 3	28–32 kHz

The three 4-kHz channels occupy a total of 12 kHz.

8.5 TDM – Time-Division Multiplexing

TDM shares a link by assigning different time slots to different sources.

Time → – A – B – C – A – B – C – A – B – C –
one repeating TDM frame

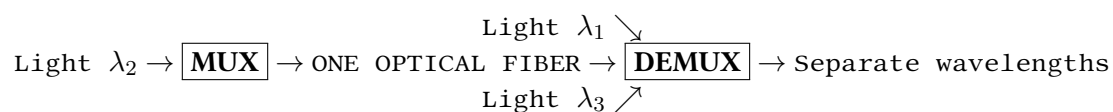
1. Inputs are divided into units.
2. Each unit receives a time slot.
3. The MUX sends the units sequentially.
4. The DEMUX uses synchronization to recover the original streams.

8.6 Synchronous vs Statistical TDM

Synchronous TDM	Slots are fixed/preassigned. A source can receive a slot even when it has no data.
Statistical / asynchronous TDM	Slots are dynamically assigned to sources that currently have data, avoiding many empty slots.

8.7 WDM – Wavelength-Division Multiplexing

WDM is used with optical fiber. Multiple optical signals are transmitted simultaneously, each using a different wavelength.



The receiver separates the wavelengths and sends the individual channels to their destinations.

8.8 WDM Transponder Concepts

The supplied material describes transponders as supporting functions such as wavelength conversion, interoperability, regeneration, and flexibility. The 3R concept is associated with **reshape, retime, and reamplify**.

8.9 FDM vs TDM vs WDM

Aspect	FDM	TDM	WDM
Division	Frequency	Time	Optical wavelength
Main medium focus	Frequency channels	Shared time slots	Optical fiber
Supplied-note emphasis	Analog	Digital	Optical
Key idea	Different frequency bands	Different time slots	Different light wavelengths

Exam Focus

10-mark answer: Define multiplexing → explain why it is needed → draw MUX/DEMUX → explain FDM, TDM and WDM → give one example/diagram → compare all three.

9. Switching Techniques

9.1 Why Switching is Needed

Large networks contain many intermediate devices and often provide multiple possible paths. Switching determines how communication progresses through the network.

9.2 Hub, Switch and Router

Device	Layer	Basic behavior
Hub	Physical Layer	Repeats/broadcasts the incoming signal to ports.
Switch	Data Link Layer	Uses link-layer addresses to forward a frame toward the appropriate port.
Router	Network Layer	Connects networks and forwards packets/datagrams using network-layer addressing/routing.

Do Not Confuse

Easy memory rule: Hub = Physical, Switch = Data Link, Router = Network.

9.3 Traditional Switching Techniques

The three major techniques are:

1. Circuit switching
2. Packet switching
3. Message switching

Packet switching can be further considered as **datagram** or **virtual-circuit** switching.

9.4 Switching by Layer

Layer	Switching described in the supplied material
Physical	Circuit switching; signals use a selected path.
Data Link	Packet switching where packet can refer to a frame/cell; VC-style operation is discussed.
Network	Packet switching; datagram or virtual circuit. The Internet is described as using datagram switching.
Application	Message switching.

9.5 Circuit Switching

Circuit switching establishes a dedicated path before data transfer.

Source $\rightarrow$ Switch 1 $\rightarrow$ Switch 2 $\rightarrow$ Destination
Dedicated reserved channel
Setup $\rightarrow$ Data Transfer $\rightarrow$ Teardown

Phases

1. **Setup** – establish the route and reserve a channel on each link.
2. **Data transfer** – data travels over the reserved path.
3. **Teardown** – release the reserved resources.

Efficiency and Delay

Resources remain reserved for the connection even when no useful data is being sent. Therefore utilization can be poor for bursty traffic. After setup, data transfer has low per-switch waiting because resources are already reserved.

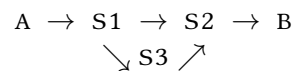
9.6 Packet Switching

A message is divided into packets. Resources are allocated on demand.

- No permanent reservation for the whole communication.
- Packets may queue at intermediate switches.
- Shared resources are used by whichever packets need them.
- Packet switching improves resource utilization but may introduce queueing delay.

9.7 Datagram Network

A datagram network is connectionless. Each packet is treated independently.



Packet 1 and Packet 2 may choose different paths.

- No setup/teardown phase.
- Switches/routers use forwarding tables.
- Each packet contains destination addressing information.
- Packets from one message can take different paths.

9.8 Virtual-Circuit Network

A virtual circuit combines a logical established path with packetized data.

Setup → Data Transfer → Teardown

All data packets follow the established logical path.

- Setup creates state in intermediate switches.
- During data transfer, packets use local VC identifiers.
- All packets belonging to the VC follow the same established path.
- Teardown removes the VC state.

9.9 Global Addressing vs Local Addressing

Global addressing	Used to establish/identify the endpoints of the virtual circuit.
Local addressing / VCI	Used during data transfer; identifier has local/switch/link scope and may change from one link to another.

9.10 Virtual-Circuit Identifier (VCI)

A VCI is a small identifier associated with a virtual circuit on a local link/switch. A switch can map an incoming VCI to an outgoing interface and a new outgoing VCI.

Incoming: Port 1, VCI 7
 $\downarrow$ **Switch table lookup**
 Outgoing: Port 3, VCI 12

Remember

VCI is local. The identifier can change at each switch. This is different from an end-to-end/global address.

9.11 Message Switching

In message switching, the complete message is treated as the forwarding unit. Each intermediate device stores the message and forwards it later.

- No dedicated circuit is reserved.
- Store-and-forward operation is central.

- Large messages can create significant storage and delay requirements.

9.12 Circuit vs Packet vs Message Switching

Feature	Circuit	Packet	Message
Forwarding unit	Continuous/circuit resource	Packet	Whole message
Resource allocation	Reserved	On demand	On demand
Path	Dedicated	Datagram may vary; VC fixed	Forwarded hop by hop
Delay	Setup delay, then low transfer waiting	Queueing possible	Potentially high store-and-forward delay
Efficiency	Lower if circuit is idle	Good resource sharing	Shared resources but large messages consume storage

Answer Skeleton

10-mark structure for switching:

1. Define switching and explain why it is required.
2. Draw a simple switched network.
3. Explain circuit switching with setup, data transfer and teardown.
4. Explain packet switching.
5. Explain datagram and virtual-circuit networks.
6. Explain VCI and global/local addressing.
7. Explain message switching.
8. Finish with a comparison table based on resource allocation, path, delay and efficiency.

10. Bandwidth, Throughput and Delay

10.1 Bandwidth

In the signal context:

$$BW = f_{\text{high}} - f_{\text{low}}$$

Bandwidth represents the frequency range occupied by a signal.

10.2 Throughput

Throughput is the effective amount of successfully transmitted data per unit time. It depends on the available capacity, traffic, protocol overhead, congestion, receiver limitations, and other conditions.

10.3 Delay

Delay is the time associated with data moving through a network. In switched networks, queueing delay can occur when packets wait at an intermediate device.

Bandwidth	Nominal frequency range/capacity concept.
Throughput	Actual successful transfer rate.
Delay	Time required for data to progress through the network.
Congestion	Excess offered traffic causing queues, loss, or increased delay.

Do Not Confuse

High bandwidth does **not** automatically mean high throughput. A high-capacity link can still suffer from congestion, queueing, protocol overhead, or receiver limitations.

11. Network Layer and IP

11.1 Network Layer Responsibilities

The Network Layer provides host-to-host delivery across an internetwork. IP defines the datagram format and logical addressing, while routers forward datagrams hop by hop.

11.2 Network-Layer Protocols

Protocol	Main role
IP / IPv4	Defines datagram format, logical addressing and forwarding.
ICMPv4	Reports network-layer delivery/routing problems.
IGMP	Supports multicast/group membership.
DHCP	Helps a host obtain a network-layer address.
ARP	Resolves an IP address to a link-layer address for local delivery.

11.3 Services Provided by the Network Layer

The Network Layer is responsible for logical addressing and forwarding/routing. IP itself is connectionless and does not provide TCP-style end-to-end flow/error/congestion control.

11.4 Routing Table

A forwarding/routing table associates destination information with a route metric and a next hop/output interface.

Destination	Metric / cost	Next hop / output
Destination network/host	Best-known route cost	Where the packet should be forwarded

11.5 IPv4 Address

IPv4 uses a 32-bit address.

$$2^{32} = 4,294,967,296$$

possible bit patterns exist in the 32-bit address space.

The supplied material emphasizes that an IP address identifies a network connection/interface rather than simply being a permanent identity of a physical device. A multi-homed device can have multiple IP addresses.

11.6 IPv4 Notation

Binary	32 bits, normally grouped into four 8-bit octets.
Dotted decimal	Four decimal octets separated by dots; each ranges from 0 to 255.
Hexadecimal	Eight hexadecimal digits; each digit represents 4 bits.

11.7 Hierarchical Addressing

An IPv4 address can be viewed hierarchically as a **network/prefix part** plus a **host/node part**.

$$[\text{Network} / \text{Prefix}] [\text{Host} / \text{Node}]$$

11.8 Classful Addressing

The historical classful scheme contains Classes A, B, C, D and E. The major unicast classes use fixed network-prefix boundaries.

Class	General role	Prefix idea
A	Large networks	Short network prefix, large host portion.
B	Medium networks	Intermediate split.
C	Smaller networks	Longer network prefix, smaller host portion.
D	Multicast	Used for multicast addressing.
E	Reserved/experimental	Historical reserved range.

11.9 Classless Addressing

Classless addressing removes fixed class boundaries and uses variable-length network prefixes/blocks. It provides more flexible allocation than the historical classful scheme.

Remember

For exam purposes, distinguish **classful** = **fixed class boundaries** from **classless** = **variable-length prefix/block**.

12. Unicast Routing

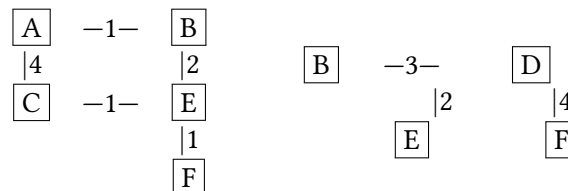
12.1 What is Unicast Routing?

Unicast means **one-to-one** delivery. A packet has one intended destination and is forwarded hop by hop toward it.

12.2 Network as a Graph

For routing algorithms, a network can be represented as a graph:

- routers are represented as nodes/vertices;
- links between routers are represented as edges;
- each edge can have a cost/metric.



12.3 Least-Cost Routing

The least-cost route is the route whose total metric is minimum among the available paths.

12.4 Least-Cost Tree

A least-cost tree has a chosen source router as its root. The path from that root to every other router is a least-cost path.

Remember

Think of a **least-cost tree** as one source router asking: “What is the cheapest route from me to everyone else?”

12.5 Distance Vector Routing

Each router maintains:

1. a destination;

2. a best-known distance/cost to that destination;
3. the outgoing line/next hop to use.

Routers exchange routing information with neighbouring routers and progressively update their tables.

Initialization

- Distance to itself = 0.
- Distance to other routers = infinity/unknown initially.
- Directly connected neighbour costs are known.

12.6 Bellman-Ford – Reference Only

The supplied equation expresses the core idea:

$$D_{xy} = \min\{c_{xa} + D_{ay}, c_{xb} + D_{by}, \dots\}$$

The important exam concept is not the arithmetic. Router x considers reaching destination y through different neighbours and chooses the neighbour that produces the minimum total cost.

Exam Focus

For your preparation: understand the Bellman-Ford idea and the table update process. Heavy mathematical derivation is not the focus of this bank.

12.7 Link-State Routing

The supplied routing material explains that distance-vector routing was replaced in the ARPANET because of slow convergence after topology changes. Link-state routing distributes topology information so that routers can calculate shortest paths themselves.

12.8 Five Steps of Link-State Routing

1. Discover neighbouring routers and their addresses.
2. Determine the cost/distance to each neighbour.
3. Construct a link-state packet containing the learned information.
4. Send/receive link-state information so routers obtain the required topology.
5. Compute shortest paths to other routers.

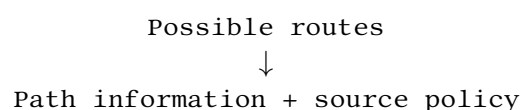
12.9 LSDB and Dijkstra

A router stores the topology information in a **Link-State Database (LSDB)**. It can then use Dijkstra's algorithm to construct a least-cost tree rooted at itself.

1. Start with the source router as the root.
2. Select the closest not-yet-added node.
3. Add that node to the tree.
4. Update the best known costs of remaining nodes.
5. Repeat until all nodes have been included.

12.10 Path Vector Routing

Path-vector routing allows route selection to be influenced by policy. The source can control the preferred path rather than simply selecting the numerically least-cost route.



↓
Selected route

12.11 Spanning Tree in Path Vector Context

In the supplied framing, the selected tree/path is determined according to source policy rather than necessarily being a pure least-cost tree.

12.12 Distance Vector vs Link State vs Path Vector

Aspect	Distance Vector	Link State	Path Vector
Information	Distance/next-hop information from neighbours	Distributed topology information	Path information plus policy
Main idea	Best distance via a neighbour	Compute shortest path from topology	Select route according to policy/path attributes
Core algorithm/concept Strength	Bellman-Ford Simple distributed approach	Dijkstra Topology-aware shortest-path computation	Policy-based path selection Strong route-policy control
Key issue	Convergence can be slow	More state/topology information required	Route choice may not be least-cost

13. RIP, IS-IS and OSPF – Brush-Up

13.1 RIP

RIP is associated with the distance-vector family. For exam memory, connect RIP with neighbour-based distance information.

13.2 IS-IS

IS-IS is associated with link-state routing. It distributes topology information and supports shortest-path computation.

13.3 OSPF

OSPF is also a link-state routing protocol. It uses topology information and shortest-path concepts.

RIP	Distance-vector family.
IS-IS	Link-state family.
OSPF	Link-state family.

Remember

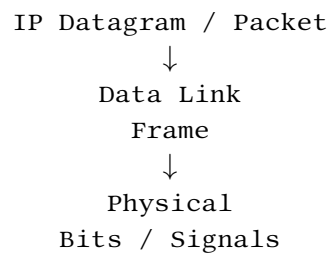
Brush-off rule: RIP = DV; IS-IS = LS; OSPF = LS. Focus your deeper preparation on the routing algorithm concepts unless a question specifically asks for protocol details.

14. Integrated Data Journey

14.1 Follow One Packet Through the Stack

This is the most useful mental model for connecting the entire syllabus.

Application
Message
↓
Transport
Segment / User Datagram
↓
Network



14.2 What Happens at a Router?

At an intermediate router:

1. The incoming physical signal becomes bits.
2. The incoming frame is processed at the Data Link Layer.
3. The network-layer datagram is examined.
4. The routing/forwarding table determines the next hop.
5. A new link-layer frame is constructed for the outgoing link.
6. The frame is transmitted again as signals.

```

Frame 1: [MAC A – IP packet – MAC R1]
          ↓ Router 1
Frame 2: [MAC R1 – IP packet – MAC R2]
          ↓ Router 2
Frame 3: [MAC R2 – IP packet – MAC B]
  
```

Do Not Confuse

The **MAC addresses change hop by hop**; the logical IP source/destination in the datagram represents the end-to-end network-layer communication.

14.3 Three Scope Ideas

Process/application	Which application or service is communicating?
IP/network layer	Which network-layer destination should receive the datagram?
MAC/link layer	Which next-hop interface should receive the current frame?

15. High-Yield Comparison Sheet

15.1 OSI vs TCP/IP

Feature	OSI	TCP/IP
Purpose	Reference model	Internet protocol suite/model
Layers	7	Common five-layer presentation in these notes
Upper layers	Application, Presentation, Session	Application grouping
Transport	Transport	Transport
Network	Network	Internet/Network
Lower layers	Data Link, Physical	Data Link, Physical

15.2 Circuit vs Packet vs Message Switching

Feature	Circuit	Packet	Message
Setup	Required	Datagram: no; VC: yes	Not a dedicated circuit setup
Unit	Reserved circuit/channel	Packet	Whole message
Resources	Reserved	On demand	On demand
Queueing	Low after setup	Possible	Store-and-forward delay
Efficiency	Lower when idle	Generally high sharing	Shared but large storage burden

15.3 Pure ALOHA vs Slotted ALOHA vs CSMA

Feature	Pure ALOHA	Slotted ALOHA	CSMA
Carrier sensing	No	No	Yes
Start time	Any time	Slot boundary	Depends on persistence method
Collision reduction	Random retransmission	Synchronization	Sense before transmit
Main limitation	Large vulnerable time	Requires synchronization	Propagation delay can still cause collisions

15.4 DV vs LS vs PV

See Section 12.12. Remember:

DV = distances from neighbours LS = topology PV = path + policy

16. Formula and Number Sheet

Topic	Formula / fact
Frequency and period	$f = 1/T$
Wavelength	$\lambda = v/f$
Bandwidth	$BW = f_{\text{high}} - f_{\text{low}}$
IPv4 address space	$2^{32} = 4,294,967,296$
Block coding	$n = k + r$
Hamming-distance detection relation	$d_{\text{min}} = s + 1$
Pure ALOHA vulnerable time	$2T_{fr}$
Pure ALOHA throughput	$S = Ge^{-2G}$
Pure ALOHA maximum	$S_{\text{max}} \approx 0.184$ at $G = 1/2$
Slotted ALOHA vulnerable time	T_{fr}
Slotted ALOHA throughput	$S = Ge^{-G}$
Slotted ALOHA maximum	$S_{\text{max}} \approx 0.368$ at $G = 1$
Bellman-Ford concept	Choose the minimum total cost through candidate neighbours.

17. Question Bank

Definitions / 2–3 marks

1. Define data communication.
2. What are the five components of data communication?
3. Define simplex, half-duplex and full-duplex communication.
4. What is a protocol?
5. What is the OSI model?
6. What is TCP/IP?
7. What is a PDU?
8. What is encapsulation?
9. What is a frame?

10. What is the role of the Physical Layer?
11. What is bandwidth?
12. What is bit rate?
13. Define guided and unguided media.
14. What is encoding?
15. What is decoding?
16. Define the Data Link Layer.
17. What is framing?
18. What is flow control?
19. What is error control?
20. What are DLC and MAC?
21. Distinguish unicast, multicast and broadcast.
22. What is ARP?
23. What is an ARP cache?
24. What is multiplexing?
25. What is FDM?
26. What is TDM?
27. What is WDM?
28. Define switching.
29. What is circuit switching?
30. What is packet switching?
31. What is a datagram network?
32. What is a virtual circuit?
33. What is a VCI?
34. What is throughput?
35. What is a routing table?
36. What is unicast routing?
37. What is distance-vector routing?
38. What is link-state routing?
39. What is path-vector routing?
40. What is an LSDB?

5-mark questions

1. Explain the OSI reference model.
2. Explain the TCP/IP protocol suite and PDUs.
3. Compare TCP and UDP.
4. Explain encapsulation and decapsulation.
5. Explain guided and unguided transmission media.
6. Explain encoding and decoding with a simple diagram.
7. Explain Data Link Layer services.
8. Explain framing and any two framing methods.
9. Explain flow control and stop-and-wait.
10. Explain Go-Back-N and Selective Repeat.
11. Explain error detection and CRC.
12. Explain the channel allocation problem.
13. Explain Pure ALOHA and vulnerable time.
14. Explain Slotted ALOHA and compare it with Pure ALOHA.
15. Explain CSMA and its persistence methods.
16. Explain CSMA/CD.
17. Explain ARP operation and ARP cache.
18. Explain multiplexing.
19. Explain FDM, TDM and WDM.
20. Explain circuit switching.
21. Explain packet switching and datagram networks.

22. Explain virtual-circuit networks and VCI.
23. Explain network-layer services and routing tables.
24. Explain distance-vector routing.
25. Explain link-state routing and Dijkstra.
26. Explain path-vector routing.

10-mark questions

1. Explain the OSI and TCP/IP models, including layer functions, protocols and PDUs.
2. Explain the Data Link Layer in detail, including services, framing, flow control and error control.
3. Explain MAC and random-access protocols from Pure ALOHA to CSMA/CD.
4. Explain ARP in detail, including its operation, cache and packet format.
5. Explain multiplexing and compare FDM, TDM and WDM with diagrams.
6. Explain circuit, packet and message switching and compare their efficiency and delay.
7. Explain virtual-circuit switching, including global addressing, local addressing, VCI and the three phases.
8. Explain the Network Layer, routing table and IPv4 addressing.
9. Explain unicast routing and compare Distance Vector, Link State and Path Vector routing.
10. Explain Link-State routing with the five steps and Dijkstra's least-cost-tree process.
11. Explain the complete journey of data through the TCP/IP stack and across an intermediate router.

18. 10-Mark Answer Templates

Question A: Explain the OSI/TCP-IP models.

[10 marks]

Answer Skeleton

Write in this order:

1. Define protocol layering.
2. Draw the OSI seven-layer model.
3. Explain the TCP/IP layers.
4. State the PDU at each layer.
5. Give important protocols such as TCP, UDP and IP.
6. Explain encapsulation and decapsulation.
7. Finish with a short OSI vs TCP/IP comparison.

Question B: Explain the Data Link Layer.

[10 marks]

Answer Skeleton

1. Define node-to-node communication.
2. Explain the frame and its header/trailer.
3. Explain DLL services.
4. Explain point-to-point and broadcast links.
5. Explain DLC and MAC sublayers.
6. Explain link-layer addressing.
7. Explain framing.
8. Explain flow control and error control.
9. Add stop-and-wait/sliding-window concepts if required.
10. Conclude with the role of DLL between Network and Physical Layers.

Question C: Explain MAC random-access protocols.**[10 marks]****Answer Skeleton**

1. Define the shared-channel/MAC problem.
2. Explain random access.
3. Explain Pure ALOHA and vulnerable time.
4. Give Pure ALOHA throughput.
5. Explain Slotted ALOHA and compare it with Pure ALOHA.
6. Explain CSMA.
7. Explain 1-persistent, non-persistent and p-persistent methods.
8. Explain CSMA/CD.
9. Mention CSMA/CA for wireless networks.

Question D: Explain ARP.**[10 marks]****Answer Skeleton**

1. Define ARP and state IP → MAC resolution.
2. Explain why a MAC address is needed for local delivery.
3. Draw ARP request and reply.
4. Explain broadcast request.
5. Explain unicast reply.
6. Explain ARP cache and timeout.
7. Draw/list the ARP packet fields.
8. Conclude with how the resolved MAC is used to construct the frame.

Question E: Explain multiplexing.**[10 marks]****Answer Skeleton**

1. Define multiplexing and explain why it is needed.
2. Draw MUX → shared link → DEMUX.
3. Explain FDM with frequency channels and guard bands.
4. Give a small FDM example.
5. Explain TDM and time slots.
6. Distinguish synchronous and statistical TDM.
7. Explain WDM and optical wavelengths.
8. Compare FDM, TDM and WDM.

Question F: Explain switching techniques.**[10 marks]****Answer Skeleton**

1. Define switching.
2. Explain circuit switching: setup, transfer, teardown.
3. Explain packet switching.
4. Explain datagram switching.
5. Explain virtual-circuit switching.
6. Explain global address and local VCI.
7. Explain VC setup, data-transfer and teardown.
8. Explain message switching.
9. Compare all three by resource allocation, path, delay and efficiency.

Question G: Explain unicast routing algorithms.**[10 marks]****Answer Skeleton**

1. Define unicast routing.
2. Explain routing tables and least-cost routing.
3. Explain Distance Vector and its table.
4. State Bellman-Ford concept without unnecessary mathematics.
5. Explain Link State and its five steps.
6. Explain LSDB and Dijkstra.
7. Explain Path Vector and policy-based route selection.
8. Compare DV, LS and PV.

19. Last-Minute Revision Page**Data communication**

Delivery + Accuracy + Timeliness + Jitter.

Data-flow modes

Simplex = one direction.

Half-duplex = both, one at a time.

Full-duplex = both simultaneously.

PDU chain

Message → Segment/User Datagram → Data-gram/Packet → Frame → Bits.

Physical

Bits + signals + transmission media.

Encoding

Bits → signal.

Decoding

Signal → bits.

DLL

Node-to-node + framing + flow/error control.

Frame

Header + network packet + trailer.

DLC

Link control.

MAC

Shared-medium access.

Addressing

IP = end-to-end logical destination.

MAC = local next-hop address.

ARP

IP → MAC.

Pure ALOHAVulnerable = $2T_{fr}$. $S = Ge^{-2G}$.Max ≈ 0.184 .**Slotted ALOHA**Vulnerable = T_{fr} . $S = Ge^{-G}$.Max ≈ 0.368 .**CSMA**

Sense before transmit.

Persistence

1-persistent = transmit immediately when idle.

Non-persistent = random wait if busy.

p-persistent = transmit with probability p in slotted channel.**CSMA/CD**

Transmit + detect collision + abort + backoff + retry.

Multiplexing

FDM = frequency.

TDM = time.

WDM = wavelength.

Switching

Circuit = reserved path.

Packet = on-demand packets.

Message = whole message store-and-forward.

Datagram

Connectionless; packets can take different paths.

Virtual circuit

Setup + data + teardown.

Local VCI can change hop by hop.

Bandwidth $BW = f_{high} - f_{low}$.**Throughput**

Actual successful data rate.

IPv4

32 bits.

 2^{32} possible patterns.

Routing table

Destination + metric + next hop/output.

Unicast

One-to-one.

Distance Vector

Neighbour distances + Bellman-Ford idea.

Link State

Topology + LSDB + Dijkstra.

Path Vector

Path information + policy.

Protocols

RIP = DV.

IS-IS = LS.

OSPF = LS.

Core memory chain

Layers tell you *where*.

PDUs tell you *what unit*.

Addresses tell you *who*.

Routing tells you *where next*.

MAC tells you *which local interface*.

Switching tells you *how the network shares paths/resources*.

Multiplexing tells you *how multiple signals share one link*.

20. Common Exam Traps

1. **Bandwidth $\neq$ throughput.** Bandwidth is capacity/frequency range; throughput is actual successful transfer rate.
2. **Frame $\neq$ packet.** Frame is the Data Link Layer PDU; packet/datagram is associated with the Network Layer.
3. **ARP direction:** IP $\rightarrow$ MAC/link address.
4. **IP vs MAC:** IP addressing is logical/end-to-end; MAC addressing is local/link-specific.
5. **Pure ALOHA vulnerable time:** $2T_{fr}$.
6. **Slotted ALOHA vulnerable time:** T_{fr} .
7. **CSMA does not eliminate collisions.** Propagation delay matters.
8. **Circuit switching reserves resources.** Packet switching allocates them on demand.
9. **VCI is local.** It can change at each switch/link.
10. **Datagram vs VC:** Datagram packets are independently routed; VC packets follow an established logical path.
11. **DV vs LS:** DV exchanges distance/next-hop information; LS distributes topology information.
12. **Path Vector:** route choice can follow policy rather than simply least cost.
13. **Multiplexing vs switching:** multiplexing shares one physical link among multiple signals; switching determines how traffic progresses through network paths.
14. **Hub/Switch/Router:** Physical/Data Link/Network.

21. Source Scope

These notes are consolidated from the supplied Networking lecture materials and the manually prepared topic checklist. The lecture PDFs are treated as the primary source for technical content; the checklist is used to ensure requested topics and personal emphasis areas are covered.

The consolidated source areas include:

- Introduction and Models – data communication, network criteria, protocol layering, OSI/TCP-IP, PDUs, encapsulation, transport/network concepts and physical-layer basics.
- Data Link Layer – node-to-node communication, DLL services, framing, flow control, error control, addressing, ARP, MAC, ALOHA, CSMA, persistence and collision handling.
- Multiplexing – MUX/DEMUX, FDM, TDM, WDM and related efficiency/capacity concepts.
- Switching – circuit, packet, datagram, virtual circuit, message switching, VCI, phases, efficiency and delay, plus hub/switch/router layer placement.
- Unicast Routing and IP – routing tables, least-cost routing/tree, Distance Vector, Bellman-Ford concept, Link State, Dijkstra, Path Vector, spanning-tree concepts and IPv4 addressing.

Remember

Study principle: understand the mechanism first, then memorize the exam wording. For long answers, use the pattern **definition** → **purpose** → **working** → **diagram** → **example/formula** → **comparison/conclusion**.